**FACULTY OF ENGINEERING TECHNOLOGY**

Subject: comm.& Data Transmission Mid Exam Fall 22/23

Name:………………………. Duration: 50 min

**THIS EXAM CONTAINS TWO QUESTIONS**

**Q1:(Please** write your answers into table only, otherwise your answers will be neglected)

| *Question* | *1* | *2* | *3* | *4* | *5* | *6* | *7* | *8* | *9* | *10* |
|---|---|---|---|---|---|---|---|---|---|---|
| *Answer* | | | | | | | | | | |
| *Question* | *11* | *12* | *13* | *14* | | | | | | |
| *Answer* | | | | | | | | | | |



1. In start stop asynchronous method the synchronization is found at:
   a) the byte level b) at the bits level c) at stop level d) at start level

2. Analog sinusoidal signal of 10v(p-p) has:
   a) An infinite number of values
   b) A definite number of values
   c) Has only one value
   d) Has only two values

3. The period of the electrical power in Jordan is equal to :
   a) 20ms b) 20µs c) 17ms d) 17 µs

4. What is the frequency of AC current which reaches zero value every 0.01 second?

   a) 100 Hz b) 50Hz c) 0 d) 500 Hz

5. A sinusoidal alternating current having frequency of 50 Hz has a peak value of 10 A. what is the value of current after 1/300 s from zero.
   a) 8.66 A b)181.5 mA c) 50 A d) 16.66 A

6. Two AC currents are given by: $i_1=50\sin(314t+\pi/6)$ and $i_2=20\sin(314t-\pi/4)$ Which of the two current leads the other?
   a) $i_1$ leads $i_2$ by $75^\circ$  b) $i_2$ leads $i_1$ by $75^\circ$
   c) $i_1$ leads $i_2$ by $15^\circ$  d) $i_2$ leads $i_1$ by $15^\circ$

7. Quantization noise in PCM, for a given input, is proportional to the :
   a) sampling rate
   b) maximum frequency of modulating signal
   c) number of digits used for coding
   d) input signal

8. In a communication system, the noise is most likely to affect the signal:
   a) at the transmitter  b) at the circuit
   c) at the channel  d) at the destination

9. The human voice signal usually is:
   a) nonperiodic, composite with limited range of frequencies
   b) periodic, un composite with limited range of frequencies
   c) nonperiodic, composite with unlimited range of frequencies
   d) periodic, un composite with unlimited range of frequencies

10. Radio station in FM broadcasting has:
   a) 200 Hz bandwidth  b) 200 kHz bandwidth
   c) 200 MHz bandwidth  d) 200 GHz bandwidth

11. An amplifier has an output of 20w.what is the output in dBw:
   a) 13 dBw  b)130 dBw  c) 13dBmw  d) 130dBmw

12. If we want to transmit 1000 character with each character encoded as 8 bits using asynchronous transmission, then the number of bits need are:
   a) 8000 bits  b) 10000 bits  c) 4000 bits  d) 5000 bits

13. Byte synchronization in synchronous transmission is accomplished by:
   a) physical layer  b) data link layer
   c) application layer  d) network layer

14. Forming bits into frames is performed by:
   a) physical layer  b) data link layer
   c) application layer  d) network layer

**Q2**: A power amplifier has an input of 90 mV across 10 KΩ. The output is 7.8 v across 8Ω speaker:

a) What is the power gain, in decibel
b) If the power amplifier is connected to two other amplifiers with gains 10 dB & 20 dB and two filters with attenuations of -2 dB & -4 dB calculate overall gain.



GOOD LUCK

| ***Question*** | ***1*** | ***2*** | ***3*** | ***4*** | ***5*** | ***6*** | ***7*** | ***8*** | ***9*** | ***10*** |
|---|---|---|---|---|---|---|---|---|---|---|
| ***Answer*** | ***B*** | ***A*** | ***A*** | ***B*** | ***A*** | ***A*** | ***C*** | ***C*** | ***A*** | ***B*** |
| ***Question*** | ***11*** | ***12*** | ***13*** | ***14*** | | | | | | |
| ***Answer*** | ***A*** | ***B*** | ***B*** | ***B*** | | | | | | |